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Populism and Corruption

Data Science Laboratory Project

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Dataset: V-Dem Country-Year Dataset (Full+Others, Version 16) and V-Party Dataset (Version 2), Varieties of Democracy Institute (V-Dem), University of Gothenburg. Source: www.v-dem.net/data

AI Declaration
AI coding agents, OpenAI Codex and Anthropic Claude Code, were used to support code development, debugging, testing, documentation, and language revision. All AI-generated outputs were reviewed, validated, and, where necessary, modified by the authors. The research design, methodological decisions, interpretation of results, and final conclusions remained under the authors' direct supervision and responsibility.

Abstract

This study examines the association, temporal direction, and institutional distribution of the relationship between governing-party populism and political corruption. V-Dem and V-Party data are analysed through descriptive comparisons, fixed-effects regressions, robustness checks, bidirectional temporal models, and a decomposition across corruption dimensions. The positive cross-country association does not persist in within-country specifications, which instead produce a small negative coefficient whose statistical significance is not robust across all models. The temporal analyses provide no consistent evidence that governing populism predicts subsequent increases in corruption or that corruption predicts later governing populism. Decomposing the outcome by institutional sphere, the contemporaneous negative association is concentrated in judicial corruption and absent in the executive dimension; a possible delayed positive effect on legislative corruption emerges in some specifications but does not survive stricter checks. Overall, the results do not support a general positive or clearly directional relationship; any association appears institutionally heterogeneous and fragile.

1 Introduction

Populist parties often frame politics as a conflict between a virtuous people and a corrupt elite [8]. This rhetoric makes corruption central to their political appeal, but the relationship may work in opposing directions. Corruption can strengthen support for anti-establishment parties, while populist governments may either challenge existing patronage networks or weaken institutional constraints once in office. Previous research has consequently examined corruption both as a source of populist mobilisation and as a possible outcome of populist politics [3].

This project assesses whether governing-party populism and political corruption are related across countries and within countries over time. It also examines their temporal ordering and whether any relationship is concentrated in a particular institutional sphere. The distinction matters because an association between more corrupt countries and more populist governments does not necessarily imply that changes in one produce changes in the other.

The report first defines the research question and its relevance. It then introduces the data and their limitations, explains the empirical strategy, and presents the association, temporal-direction, and corruption-dimension analyses. The final section summarises the evidence and discusses its limitations and implications.

2 Research Question

2.1 Statement and Scope

The project examines whether governing-party populism and political corruption are systematically related across countries and over time. The main research question is:

> *Do countries governed by more populist parties exhibit higher political corruption, which dimension of corruption is most affected, and in which temporal direction does the relationship run?*

This broad question is divided into three connected sub-questions:

1. Is governing-party populism associated with political corruption after persistent differences between countries and common changes across years are taken into account?

2. Does governing-party populism precede subsequent changes in corruption, or does corruption instead precede subsequent changes in governing-party populism?

3. Is the relationship concentrated in a specific institutional dimension of corruption, namely the executive, legislative, judicial or public sector?

The first sub-question distinguishes a cross-sectional comparison from a within-country comparison. A positive association across countries would indicate that countries governed by more populist parties also tend to display higher corruption on average. It would not, however, show that corruption changes when the populism of the governing party changes within the same country. The within-country question is therefore central to the study because it reduces the influence of stable national characteristics, such as historical institutions or long-standing political culture. Nevertheless, this comparison remains observational and cannot remove every time-varying confounder.

The second sub-question concerns temporal ordering. Two competing mechanisms are considered. Under the *populism-to-corruption* mechanism, a rise in governing-party populism may be followed by higher corruption if political power becomes more concentrated or institutional oversight is weakened. Under the *corruption-to-populism* mechanism, dissatisfaction with corrupt political elites may increase support for anti-establishment parties and make a more populist government electorally likely. The analysis consequently treats the two directions symmetrically rather than assuming in advance that populism is the cause.

The third sub-question recognises that political corruption is not a single institutional phenomenon. An aggregate index may conceal different patterns across executive office, legislatures, courts, and the public administration. The dimensional analysis is therefore intended to determine whether any observed relationship is broad or concentrated in a particular sphere.

2.2 Operational Definitions

In this report, *populism* is understood as a political orientation that presents society as divided between a virtuous people and a self-serving elite, while claiming that politics should express the general will of the people [8]. Empirically, it is measured using the V-Party populism index, which combines party-level measures of anti-elitism and people-centrism [6]. The analysis focuses on the senior governing party because the substantive question concerns populism exercised from a position of governmental power. Populism is retained as a continuous score rather than converted into a binary classification, thereby avoiding an arbitrary threshold between populist and non-populist governments.

Political corruption refers to the misuse of political or public authority for private or particularistic benefit. The primary outcome is the V-Dem Political Corruption Index, a composite measure covering corruption in the executive, legislative, judicial, and public-sector spheres [2]. The index is coded from lower to higher corruption, so a positive estimated association with governing-party populism indicates more corruption. The dimensional analysis uses the corresponding V-Dem measures for each institutional sphere, with their orientations harmonised where necessary.

The term *association* denotes statistical co-movement and does not imply causality. *Temporal direction* refers to whether past values of one variable provide information about later values of the other after observed controls, fixed effects, and persistence are considered. Lead–lag and Granger-style evidence can establish predictive precedence under a model specification, but it cannot by itself demonstrate a causal effect. Accordingly, the study uses causal language only when describing theoretical mechanisms, while interpreting the empirical results as associational or predictive evidence.

2.3 Motivation and Stakeholders

The relationship between populism and corruption is substantively important because anti-corruption appeals are often central to populist political competition. Populist parties may gain support by portraying established elites as corrupt and unresponsive, yet governing parties may themselves face incentives to weaken institutional constraints or favour political allies. This tension makes it insufficient to compare average corruption levels across countries. Evidence must also show whether changes occur within countries and whether they follow, rather than precede, changes in governing-party populism.

Several groups may benefit from the findings. Voters and journalists can use the evidence to assess whether anti-corruption claims are consistent with governments' observed institutional record. Anti-corruption agencies, judicial bodies and election-monitoring organisations may use the results to identify whether risks are associated with the arrival of more populist governments or instead form part of the political conditions that precede their electoral success. Policymakers and civil-society organisations concerned with democratic accountability may also benefit from understanding whether the relationship differs according to the strength of electoral democracy or across institutional spheres.

The study does not aim to classify populist governments as inherently corrupt or inherently anti-corrupt. Its objective is narrower: to assess whether the available cross-national panel data provide robust evidence of an average association, a temporal ordering, or a dimension-specific pattern. A null or inconclusive result remains informative because it indicates that a simple general claim is not supported by the available evidence and that more specific institutional mechanisms or research designs may be required.

3 The Data

3.1 Data Description

The study combines two datasets produced by the Varieties of Democracy (V-Dem) Institute. The V-Dem Country-Year Dataset, Full+Others, Version 16 provides annual country-level measures of political corruption, democratic institutions, and socioeconomic conditions [2, 11]. The raw file contains 28,092 country-year observations spanning 1789–2025. Within the period relevant to this study, 1970–2019, it contains 8,454 observations covering 182 countries.

The V-Party Country-Party-Date Dataset, Version 2 records political parties at specific electoral dates and includes information on party identity, ideology, populism, electoral support, and government status [6, 10]. Its raw file contains 11,898 observations for 178 countries between 1900 and 2019. The expert-coded party variables used in this project are available from 1970 onwards; the corresponding 1970–2019 subset contains 7,518 party-date observations.

The analytical dataset brings the relevant information from these two sources to a common country-year unit. It is an unbalanced panel containing 3,965 observations, 31 variables, and 96 countries over the period 1970–2019. This is the dataset used in the main statistical analyses. The procedures used to obtain it from the raw sources are discussed in Section 4.

3.2 Dataset and Variables

Table 1 summarises the principal analytical variables. Governing-party populism and the main corruption indices range from 0 to 1. Higher values indicate, respectively, greater populism and greater corruption.

Table 1: Main variables in the analytical panel

Variable	Source	Description
`populism_governing`	V-Party	Continuous populism score of the senior governing party; primary explanatory variable.
`v2x_corr`	V-Dem	Composite political corruption index; primary outcome.
`v2x_execorr`	V-Dem	Executive corruption index.
`v2x_pubcorr`	V-Dem	Public-sector corruption index.
`v2lgcrrpt`, `v2jucorrdc`	V-Dem	Legislative and judicial corruption measures.
`log_gdppc`	V-Dem	Natural logarithm of GDP per capita; control for economic development.
`v2x_polyarchy`	V-Dem	Electoral democracy index; control for democratic institutions.

The analytical panel also retains contextual variables that support interpretation and robustness checks. `party_names` identifies the senior governing party or parties; `is_election_year` distinguishes election years from other annual observations; and `years_since_last_election` records the distance from the most recent election. `e_regiongeo` provides a geographical-region classification, while `e_pop` reports population and `e_gdppc` reports GDP per capita. The related `log_gdppc` variable is the logarithmic GDP-per-capita measure used in the main models. Country identifiers, country names, and calendar year complete the core descriptive information. Variables created specifically for individual analyses are introduced in the methodology section rather than listed here.

3.3 Strengths and Weaknesses

The main strength of the data is their joint substantive and temporal coverage. The combination of country-year corruption measures with party-level populism scores permits comparisons across countries, within countries over time, and across different institutional dimensions of corruption. The common V-Dem framework also provides consistent identifiers and related measures of economic development and electoral democracy.

Several limitations affect the interpretation of the analysis. First, populism and corruption are largely expert-coded constructs and may contain measurement error or shared assessment bias. Second, V-Party observations are concentrated around elections and are less temporally dense than the annual V-Dem records. Third, the coder-count filter (excluding observations with fewer than three coders) reduces country coverage from 163 countries with governing-party data to 96 in the final panel. This selection may improve measurement reliability but limits geographical representativeness, particularly for smaller and less-documented countries. Coverage also ends in 2019, excluding more recent political developments. Finally, some institutional corruption measures, particularly the legislative and judicial indicators, have limited within-country variation and the legislative measure is unavailable where no legislature exists. These features make the data suitable for comparative panel analysis, but they require caution when interpreting small or statistically imprecise relationships.

4 Methodology

The study followed a sequential pipeline. The two raw datasets were first stored in a reproducible database and explored separately. They were then transformed into a common country-year panel. The statistical analysis proceeded from descriptive evidence to contemporaneous association, temporal direction, and finally a decomposition across corruption dimensions. This order was chosen to keep three questions distinct: whether populism and corruption differ across countries, whether they move together within the same country, and whether one systematically precedes the other.

4.1 Raw Data Acquisition and Storage

The V-Dem Country-Year and V-Party Country-Party-Date CSV files were downloaded from their official pages [2, 6]. The files were preserved unchanged and imported into a local DuckDB database through Docker. The database contains separate raw tables for V-Dem country-years and V-Party party-dates, together with metadata views used to verify row counts, country coverage, and year ranges.

DuckDB provided reproducible SQL access to both large CSV files without requiring a separate database server [12]. Docker ensured that all team members used the same database environment. A distributed or cloud platform was not adopted because the data fit comfortably on a local machine and the additional infrastructure would not have improved the analysis.

4.2 Exploratory Data Analysis

The exploratory analysis was conducted on the raw V-Dem and V-Party tables before constructing the final modelling panel. It examined temporal coverage, missing values, variable distributions, annual trends, and regional differences. Rankings of extreme values were used as basic checks that the populism and corruption measures behaved according to their documented orientation.

Populism was retained as a continuous variable because its distribution did not provide a natural binary threshold. Corruption was explored both through the composite index and through its executive, public-sector, legislative and judicial components. Selected country trajectories were also inspected around changes in government. These descriptive analyses motivated the panel models but were not interpreted causally because they could not separate cross-country differences from within-country change.

4.3 Preprocessing

Preprocessing converted the raw party-date and country-year datasets into the analytical panel described in Section 3. The study period was restricted to 1970–2019, when the required V-Party expert-coded variables are available. Parties classified as senior governing partners through `v2pagovsup` were selected and observations based on three or fewer coders for either component of the populism index were excluded. When several senior governing parties were observed in the same country-year, their populism scores were combined using vote shares when valid weights were available and a simple mean otherwise.

Because V-Party mainly records election years while V-Dem is annual, the latest observed governing-party score was carried forward until the next election. Of the 3,965 country-years in the final panel, 945 (23.8%) are election-year observations and 3,020 (76.2%) carry a forward-filled value. The median distance from the most recent election is 2 years; 93.2% of observations fall within 5 years of an election, although some extreme cases reach 43 years. The coder-count filter reduces country coverage from 163 countries with governing-party data to 96 in the final panel; this selection may improve measurement reliability but limits geographical representativeness, particularly for smaller and less-documented countries. The resulting annual series was joined to V-Dem through `country_id` and `year`. The pipeline checked required columns, duplicate country-years, annual ordering, variable ranges and corruption orientation. Missing values were not imputed; each model used complete observations for the variables required by its specification.

Feature engineering was limited to variables required by the research design. `populism_governing` represents the continuous populism score of the senior governing party. `is_election_year` and `years_since_last_election` record the electoral origin of each annual value. GDP per capita was log-transformed as `log_gdppc`. The legislative and judicial corruption measures were reoriented where necessary so that positive model coefficients consistently represent greater corruption. Finally, country-specific lags, leads, and first differences were generated for the temporal and placebo analyses. Automated high-dimensional feature generation was not used because every feature was tied to a pre-specified substantive question.

4.4 Modelling the Association

The association analysis asked whether corruption changes within a country when its governing party becomes more or less populist. The share of populism variation occurring within countries was examined first, since fixed-effects models identify the relationship from this variation. A separate persistence model assessed how strongly current corruption depends on its previous-year level.

The principal model was a two-way fixed-effects regression:

$$Corr_{it} = \beta Pop_{it} + \gamma_1 \log(GDPpc_{it}) + \gamma_2 Dem_{it} + \alpha_i + \lambda_t + \varepsilon_{it}. \tag{1}$$

Here, i identifies the country and t the calendar year. $Corr_{it}$ is the composite political corruption index, Pop_{it} is governing-party populism, $\log(GDPpc_{it})$ is log GDP per capita, and Dem_{it} is electoral democracy. The coefficient β measures their conditional within-country association, while γ_1 and γ_2 are the control coefficients. α_i represents country fixed effects, λ_t year fixed effects, and ε_{it} the remaining error.

Country effects remove stable differences between countries, while year effects absorb shocks shared across countries. This design was preferred to pooled OLS, which mixes within-country and between-country variation. It was also preferred to random effects because that model assumes that unobserved country characteristics are unrelated to the regressors [13]. A Hausman test was retained as a supporting diagnostic [5].

Three nested specifications evaluated whether the populism coefficient changed after adding log GDP per capita and electoral democracy. These controls were selected because economic development and democratic institutions can influence both corruption and populist electoral success. Additional institutional controls were not included in the primary model because some may be affected by populist government and therefore introduce over-control. Standard errors were clustered by country. First differences, alternative covariance estimators, alternative measures of governing populism, and a populism–democracy interaction were used as robustness checks.

XGBoost provided a separate predictive check for nonlinearities and interactions [1]. Entire countries were held out during grouped cross-validation, and SHAP values were used to describe feature importance [7]. This model was not used to replace fixed effects: cross-country prediction and within-country inference answer different questions, and predictive importance does not imply statistical significance or causality.

4.5 Causal-Direction Analysis

Temporal ordering was examined in both directions. The lead–lag specifications were:

$$Corr_{it} = \sum_{k=1}^{K} \beta_k Pop_{i,t-k} + \boldsymbol{\gamma}'\mathbf{X}_{it} + \alpha_i + \lambda_t + \varepsilon_{it}, \tag{2}$$

$$Pop_{it} = \sum_{k=1}^{K} \theta_k Corr_{i,t-k} + \boldsymbol{\delta}'\mathbf{X}_{it} + \alpha_i + \lambda_t + \nu_{it}. \tag{3}$$

As before, i identifies countries and t years. The index k denotes the lag length, while K is the maximum number of annual lags. $Pop_{i,t-k}$ and $Corr_{i,t-k}$ are past values of populism and corruption. β_k and θ_k are their lag-specific coefficients. $\mathbf{X}_{it}$ contains log GDP per capita and electoral democracy, with coefficient vectors $\boldsymbol{\gamma}$ and $\boldsymbol{\delta}$. Country and year effects are denoted by α_i and λ_t, while ε_{it} and ν_{it} are the respective errors.

The main specification used one- and two-year lags; windows up to five years were inspected as approximate short- and medium-term diagnostics. Since both variables are persistent, the stricter Granger-style models also included the dependent variable's own lags. Joint Wald tests assessed whether the proposed-cause lags added predictive information beyond that history. Granger-style evidence was interpreted as temporal predictability, not structural causality [4].

Future values of the proposed cause were added as placebo leads in both directions. Significant future terms would indicate anticipation, pre-trends, or omitted time-varying factors. Event studies complemented these models by examining corruption around substantial entries into populist government and populism around large corruption increases. First differences and samples restricted to elections, government changes, or observations close to elections assessed whether the findings depended on persistent levels or forward-filled values. Interactions and split samples were used to examine differences between lower- and higher-democracy contexts.

Instrumental-variable and quasi-experimental models were not adopted because the available data do not provide a defensible external instrument or clearly exogenous assignment of populist government. The analysis therefore combines multiple directional diagnostics without claiming that temporal precedence alone establishes causality.

4.6 Corruption-Dimension Decomposition

The final analytical step examined whether the composite result concealed different relationships for executive, public-sector, legislative, and judicial corruption. The composite index remained as a benchmark. Executive and public-sector indices already increase with corruption, whereas the continuous legislative and judicial estimates were multiplied by -1 so that all model outcomes had the same interpretation. Their five-level ordinal versions were used only for descriptive and measurement-robustness checks because the continuous estimates preserve more variation.

For each corruption dimension d, the controlled association model was:

$$Corr_{it}^{(d)} = \beta_d Pop_{it} + \boldsymbol{\gamma}_d' \mathbf{X}_{it} + \alpha_{i,d} + \lambda_{t,d} + \varepsilon_{it}^{(d)}. \tag{4}$$

The superscript d identifies the corruption dimension. $Corr_{it}^{(d)}$ is dimension d for country i in year t, β_d is its populism coefficient, and $\mathbf{X}_{it}$ contains log GDP per capita and electoral democracy. $\boldsymbol{\gamma}_d$ contains the corresponding control coefficients; $\alpha_{i,d}$ and $\lambda_{t,d}$ are dimension-specific country and year effects; and $\varepsilon_{it}^{(d)}$ is the error. Because the outcomes use different scales, coefficients were compared using the within-standardised effect:

$$\widetilde{\beta}_d = \beta_d \frac{SD_{\text{within}}(Pop)}{SD_{\text{within}}(Corr^{(d)})}. \tag{5}$$

Here, $SD_{\text{within}}(\cdot)$ is the standard deviation after removing each country's mean. Models were estimated on both outcome-specific samples and a common complete-case sample to separate substantive differences from sample composition.

The two directional models were then repeated for every dimension using two annual lags, dependent-variable lags, country and year effects, and joint Wald tests. Placebo leads, first differences, democracy interactions, electoral samples, alternative covariance estimators, and alternative measurement scales provided pre-declared robustness checks. Since four related subdimensions were tested repeatedly, Benjamini–Hochberg false-discovery-rate correction was applied to the association family and to the eight dimension-by-direction Granger tests. The composite was treated as a benchmark rather than an additional independent test. This confirmatory design was preferred to searching across dimensions and specifications for isolated significant coefficients.

4.7 Tools and Motivation

Python was used for the complete analytical workflow. Polars and pandas handled tabular data; DuckDB provided SQL storage and querying; NumPy and SciPy supported numerical operations and statistical tests; `linearmodels` implemented panel regressions; scikit-learn supported cross-validation; XGBoost and SHAP provided the nonlinear predictive check; and Matplotlib and Seaborn produced the figures. Jupyter notebooks connected the reusable code to the narrative analysis, while `pytest` supported testing and code quality.

The primary methods were selected according to the question rather than by predictive fit. Fixed effects addressed within-country association, lead–lag and Granger-style models examined temporal ordering, and the decomposition applied the same design consistently across institutional outcomes. Pooled regression, arbitrary binary populism thresholds, broad automated feature selection and model selection based only on fit were avoided because they would not isolate the intended comparisons. More causal designs were not used because no credible source of exogenous variation was available.

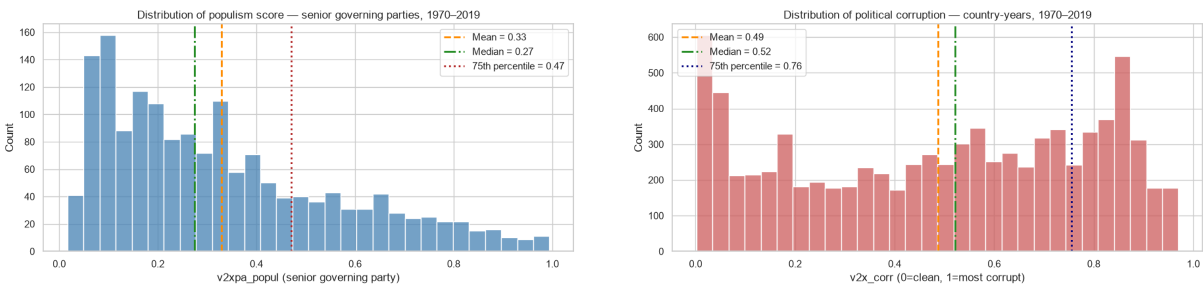
Docker made the DuckDB environment reproducible across computers. Git and GitHub supported collaborative development through separate branches, pull requests, and review before integration [9]. This organisation allowed each task to be developed independently while preserving one reproducible pipeline and a shared analytical dataset.

5 Results

5.1 Exploratory data analysis

The source-level exploratory analysis covers **1970–2019** and keeps the two datasets distinct. V-Dem contributes 8 454 country-years for 182 countries, with `v2x_corr` observed in 8 388 cases (99.2%). V-Party contributes 7 518 party-date rows for 178 countries, of which 6 307 have a non-missing party-level populism score. The governing-party distribution below is narrower: it uses 1 629 senior-governing-party rows, corresponding to 1 628 country-years after duplicate aggregation. The stricter regression panel described in Section 3 contains 3 965 country-years and 96 countries. Coverage ends in 2019, so conclusions do not extend to more recent governments. Four exploratory observations motivate the main modelling choices used in the subsequent analyses.

Two variables, two shapes. Governing-party populism is **unimodal and right-skewed** (mean 0.33, median 0.27, 75th percentile 0.47), with no visible break that would justify a binary populist/non-populist classification. Composite corruption is **bimodal**: a dense cluster of consolidated democracies near 0 (anchored by Denmark at $\texttt{v2x\_corr} = 0.002$) and a second cluster of corrupt states near 0.85 (Equatorial Guinea, Venezuela, DR Congo, Madagascar, Chad at ≈ 0.97), with comparatively few country-years in between (Figure 1). This contrast is a key observation from the exploratory analysis: populism is retained as a continuous measure, while the bimodal corruption distribution suggests substantial cross-country heterogeneity associated with regional and institutional differences. Country fixed effects subsequently remove persistent country-level differences from the within-country analysis.



(a) Senior governing-party populism (`v2xpa_popul`, $N = 1\,629$ party rows).

(b) Corruption (`v2x_corr`, $N = 8\,388$).

Figure 1: Two key variables, two different shapes. Populism is a smooth right-skewed continuum; corruption is bimodal, with a low-corruption cluster near 0 and a high-corruption cluster near 0.85.

Region dominates. Mean corruption ranges from ≈ 0.05 in Western Europe and North America to ≈ 0.85 in Central Asia (Figure 2), a cross-regional gap roughly an order of magnitude larger than the high- versus lower-populism difference reported below. The observed country rankings are consistent with the documented orientation of the index: Nordic democracies anchor the low-corruption end, while fragile and post-conflict states anchor the high-corruption end. These regional differences motivate adjustment for economic development and electoral democracy. Country-clustered standard errors instead account for serial dependence and heteroskedasticity within countries over time.

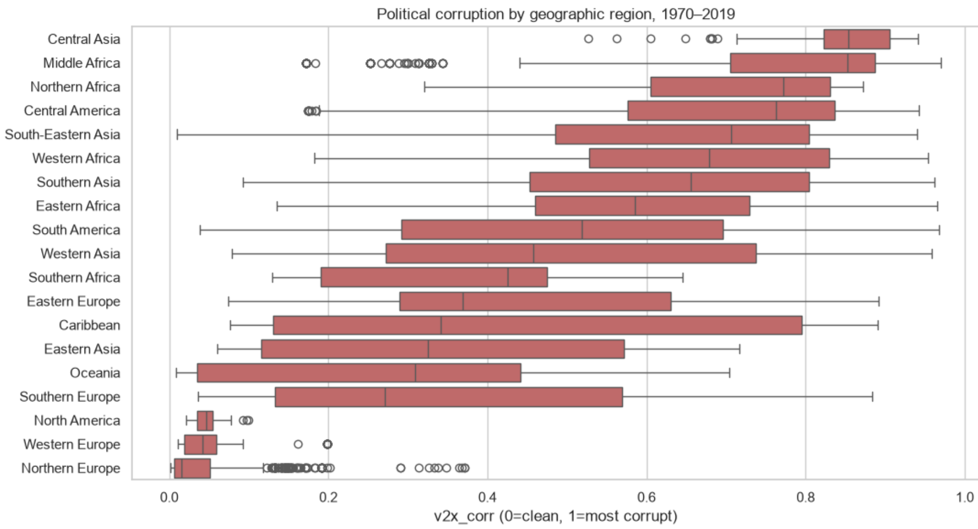


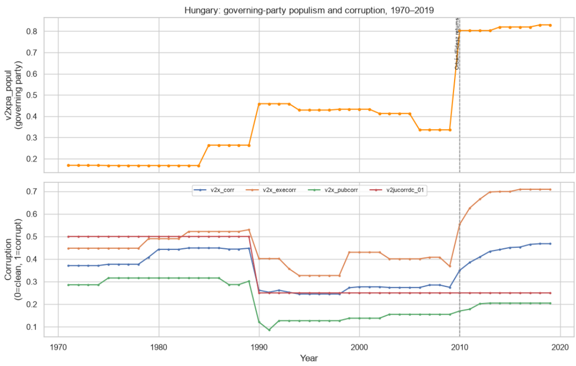
Figure 2: Composite corruption by UN-geoscheme region, 1970–2019. The cross-regional gap is roughly an order of magnitude larger than any populist–non-populist difference.

Populist governments are descriptively more corrupt, but only marginally. Using the top quartile of governing-party populism (> 0.47, descriptive only) as a split, all five corruption dimensions are higher in high-populism country-years and every difference is statistically significant at $p < 0.001$ (Mann–Whitney U; Table 2). However, every Cliff's δ falls in the *small* range (0.054–0.108): significance is driven by the large sample (1 831 vs. 5 078 country-years), not by large differences. Legislative ($\delta = 0.108$) and judicial corruption ($\delta = 0.087$) show the largest gaps; executive ($\delta = 0.054$) and public-sector corruption ($\delta = 0.057$) the smallest: a candidate mechanism pattern that the per-sphere decomposition in Section 5.5 re-tests with controls. Because the same countries appear repeatedly, these tests are descriptive and do not adjust for within-country dependence. Formal inference therefore relies on the subsequent panel models.

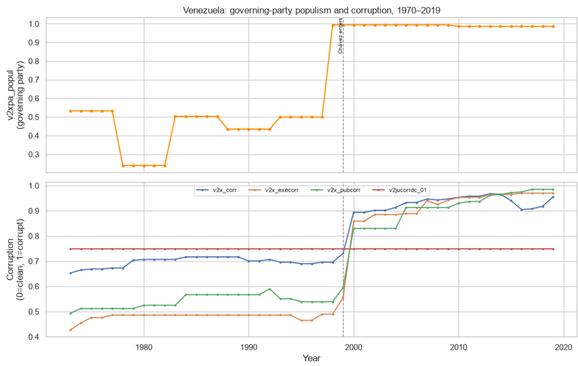
Table 2: Unadjusted descriptive Mann–Whitney U comparisons, high-populism vs. lower-populism country-years (1970–2019). The tests do not adjust for repeated observations within countries; all effect sizes are small.

Variable	N_{high}	N_{low}	p-value	Cliff's δ
`v2lgcrrpt_01`	1 648	4 848	8.4×10^{-12}	0.108
`v2jucorrdc_01`	1 831	5 078	6.9×10^{-9}	0.087
`v2x_corr`	1 831	5 078	5.6×10^{-5}	0.063
`v2x_pubcorr`	1 831	5 078	2.7×10^{-4}	0.057
`v2x_execorr`	1 831	5 078	5.9×10^{-4}	0.054

Five illustrative transitions, five different trajectories. Comparing the five-year mean of `v2x_corr` immediately before and after the entry of a well-documented populist government yields no common pattern: Venezuela rises sharply ($0.69 \rightarrow 0.87$, $+0.17$) and Hungary rises notably from a comparatively clean baseline ($0.28 \rightarrow 0.40$, $+0.13$); Turkey moves only slightly ($+0.03$); Bolivia is essentially flat ($0.70 \rightarrow 0.70$); Ecuador in fact declines slightly (-0.04). Pre-existing corruption level matters as much as the transition itself: Bolivia, Ecuador, and Venezuela were already in the upper half of the distribution before their populist leaders took office. Figure 3 contrasts the two clearest cases (Hungary, rising from a clean baseline; Venezuela, rising further from an already-corrupt one). The cases were selected to illustrate documented government transitions within the observation window; they are not a representative sample and are not used for formal inference. The broader set therefore illustrates, rather than proves, the absence of a uniform post-transition trajectory.



(a) Hungary, Fidesz/Orbán enters 2010.



(b) Venezuela, Chávez enters 1999.

Figure 3: Two contrasting populist transitions: corruption rises in both cases but from very different baselines. The full five-case set shows no common trajectory.

Taken together, the four observations above motivate the modelling design used in the remaining analyses: **continuous populism** (no natural break point); **country and year fixed effects** (the bulk of cross-sectional variation is regional, not populist); **GDP and democracy controls** (to adjust for observed differences in economic development and electoral democracy); and **country-clustered standard errors** (the panel structure makes within-country serial correlation plausible). The descriptive evidence alone cannot distinguish "populism raises corruption", "corruption breeds populism", and "both reflect a shared confounder". This uncertainty motivates the within-country regression (Section 5.3), the lead-lag direction tests (Section 5.4), and the per-sphere decomposition (Section 5.5) that follow.

5.2 Preprocessing

The assembled panel covers 96 countries over 1970–2019 (3 965 country-years). Cross-sectional Pearson and Spearman correlations between governing-party populism and the five corruption dimensions (Figure 4) are positive across the board once `v2lgcrrpt` and `v2jucorrdc` are read in the "higher = cleaner" direction. All coefficients fall in the small range ($|r| \approx 0.10$–0.13), with judicial corruption showing the strongest association ($r \approx -0.13$, i.e. $+0.13$ in the more-corrupt direction) and public-sector the weakest. The similar Pearson and Spearman coefficients suggest that the descriptive association is not highly sensitive to the rank transformation. These pooled correlations cannot separate within-country signal from regional confounding, which is the role of the fixed-effects regression.

5.3 Within-country association

Between vs. within: the sign flips. The naive pooled correlation between governing-party populism and `v2x_corr` is $+0.14$ (matching the EDA); the between-country estimator yields $+0.23$. Once country and year fixed effects strip out level differences, the sign reverses: two-way fixed effects gives -0.068. The hypothesised positive within-country association never appears — the $+0.14$ is driven by between-country differences, not within-country change. With dependence-adjusted standard errors, however, none of the four single-predictor estimates is

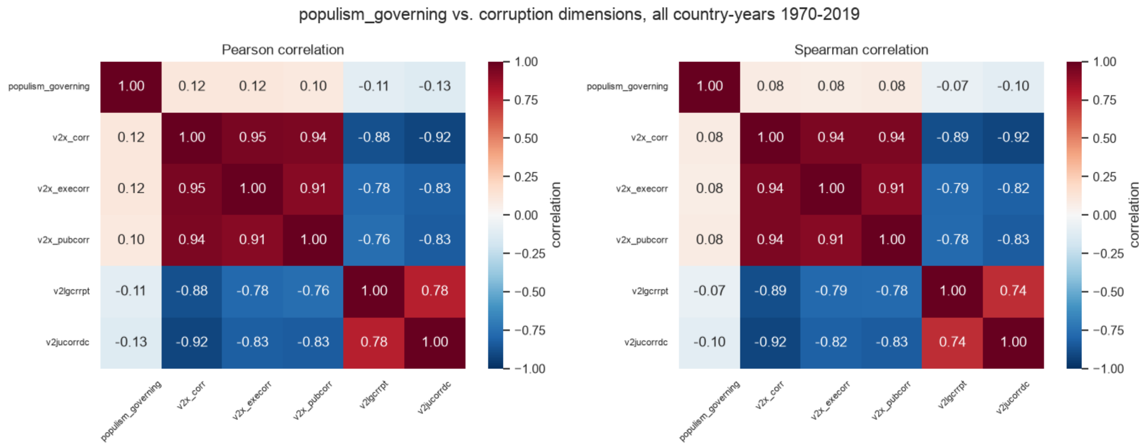


Figure 4: Pooled Pearson (left) and Spearman (right) correlations across all country-years (1970–2019). `v2lgcrrpt` and `v2jucorrdc` retain V-Dem's original direction (higher = cleaner); negative correlations there have the same substantive meaning as positive ones for the `v2x_*` indices.

itself statistically significant (pooled $p \approx 0.09$, between $p \approx 0.13$); the within-negative reaches conventional significance only once GDP and democracy controls enter.

M1–M4. The TWFE populism coefficient is -0.068 ($p = 0.064$) in M1 (populism only), -0.072 ($p = 0.043$) in M2 (+ log GDP), and -0.064 ($p = 0.036$) in M3 (+ polyarchy), essentially flat across the three controlled specifications (Figure 5). M4 adds a one-year corruption lag; after conditioning on lagged corruption, the populism coefficient falls to -0.003. This dynamic specification estimates a short-run association and is not directly comparable with the static M1–M3 estimates.

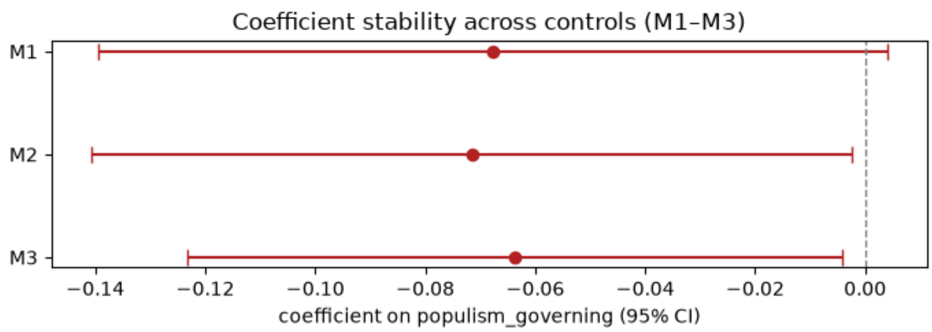


Figure 5: Coefficient stability across M1–M3. Point estimate on `populism_governing` barely moves as log GDP and polyarchy controls are added; intervals exclude zero in M2 and M3.

Sign-robust, significance-fragile. Three standard-error variants (cluster-by-country, two-way cluster, Driscoll–Kraay) return the identical point estimate (-0.072); only the interval width changes. A first-difference estimator gives -0.024 ($p = 0.20$): same sign, no longer significant. Alternative populism aggregations (whole coalition, full parliament) and the originally-specified controls (rule of law + regime type) preserve the negative sign but drop significance ($p \approx 0.11$–0.26). Across every variant the **sign is negative; the hypothesised positive never appears.**

The slope changes with democracy. Adding a `populism` $\times$ `polyarchy` interaction to M2 is significant ($+0.257$, $p = 0.012$). The model-implied marginal slope of populism on corruption is −0.16 at low polyarchy (10th percentile, significant), −0.08 at the median (significant), and ≈ +0.03 at high polyarchy (90th percentile, not significant). The estimated negative association is concentrated at lower levels of electoral democracy, although this heterogeneity result remains observational and specification-dependent.

Magnitude. A within-country SD move in populism (0.16) shifts predicted corruption by ≈ 0.010, about 3% of the overall corruption SD or 10% of the within-country corruption SD (the appropriate denominator for a within estimator). The estimated magnitude is small relative to total cross-country variation, but more noticeable relative to within-country variation. Its statistical robustness nevertheless remains limited.

Predictive check (GroupKFold XGBoost). Holding out entire countries (folds stratified by world region), an XGBoost model reaches an out-of-fold $R^2 = 0.45$ (untuned, 0.51 nested-CV tuned), versus 0.38 for a linear benchmark on the same folds. SHAP importance places `populism_governing` **last** among the three predictors (mean |SHAP|: `v2x_polyarchy` 0.156, `log_gdppc` 0.060, `populism_governing` 0.036). Because XGBoost uses raw features it can exploit between-country signal that fixed effects remove, so this is a separate cross-country prediction question, not a confirmation of the within-country result, but the consistent message is that populism is the weakest of the three signals in predicting corruption.

All of the above are **same-year associations within country**, not causal effects: whether populism leads, lags, or coincides with corruption is the question taken up in Section 5.4.

5.4 Temporal direction

A strengthened lead-lag pipeline tests both temporal directions between governing populism and `v2x_corr`: lead-lag baseline, Granger-style joint tests, first-differencing, electoral-sample robustness, event-studies, and democracy heterogeneity. Some negative coefficients emerge in the simpler lead–lag specifications in both directions. However, they are not stable across lag structures, first differences, electoral samples, and Granger-style tests (Table 3). Granger-style joint tests, which control for the outcome's own persistence, do not reject the null in either direction (populism → corruption $p = 0.69$; corruption → populism $p = 0.24$). First-differencing and the election-years-only restriction collapse the lag-1 coefficient to ≈ 0, suggesting the −0.048 baseline appears to be driven mainly by forward-filled observations. Around populist-government entry, the post-event corruption response is not significant (joint $p = 0.23$). Around corruption shocks, populism is elevated both before and after the shock (pre-trend $p = 0.08$, post $p = 0.02$): best read as clustering around the same political episode, not a clean shock-then-response sequence. The polyarchy interaction that mattered contemporaneously (Section 5.3) does not carry over to the lagged version (low-polyarchy slope $p = 0.15$). **The panel evidence shows no robust temporal direction in either direction.**

5.5 Per-sphere decomposition

The decomposition provides no evidence of a uniform populism–corruption relationship across institutions. The composite contemporaneous negative (Section 5.3) is concentrated almost entirely in **judicial corruption**, the only FDR-significant subtype (Table 4); the executive coefficient is essentially zero, while public-sector and legislative are negative but imprecise after FDR correction. The judicial result is robust to common-sample estimation ($\beta_{\text{std}} = -0.144$,

Table 3: Lag-1 coefficient on populism in a fixed-effects model for `v2x_corr` (country and year FE, log GDP and polyarchy controls). The reverse direction (corruption → populism) produces some negative coefficients in simpler specifications but is not stable across lag structures or Granger-style tests; the joint test does not reject the null in either direction ($p = 0.69$, $p = 0.24$).

Specification (populism → corruption)	Coefficient	p-value
Lead-lag baseline	−0.048	0.031
First-difference	−0.003	0.66
Election years only	−0.010	0.80
Low-polyarchy marginal slope (10th pct)	−0.089	0.15

$q = 0.042$) and to country, two-way, and Driscoll–Kraay SE variants, but weakens under first-differencing ($p = 0.033$ raw, $q = 0.134$ after FDR). The judicial slope also mirrors the composite democracy heterogeneity: steepest at low polyarchy, ≈ 0 at high. Temporally, none of the eight subtype $\times$ direction Granger-style tests survives FDR except **populism → legislative corruption**, driven by a small positive two-year coefficient ($+0.073$, $p = 0.004$, joint $q = 0.028$). The signal is **specific but fragile**: no FDR placebo warning, but first-differencing returns null ($p = 0.37$) and election-only restrictions render the estimate imprecise, best read as suggestive evidence of a delayed institutional channel, not robust causal proof.

Table 4: Contemporaneous M3 (two-way FE, log GDP and polyarchy controls) by corruption dimension, within-standardised coefficients on `populism_governing`. FDR (Benjamini–Hochberg) is applied across the four subtype tests; the composite is a benchmark, not in the FDR family ($\beta = -0.097$, $p = 0.036$).

Corruption dimension	β_{std}	q (FDR)
Executive	+0.021	0.685
Public sector	−0.097	0.113
Legislative	−0.076	0.213
Judicial	**−0.173**	**0.009**

6 Conclusion

The aim of this project was to assess whether governing-party populism is associated with political corruption, whether the relationship has a clear temporal direction, and whether it differs across institutional dimensions. The evidence only partially closes the gap between this goal and the results that can be obtained from the available data.

Addressing the three sub-questions in turn: first, descriptive comparisons show that high-populism country-years tend to be more corrupt, but the effect sizes are small and the pattern is strongly shaped by cross-country and regional differences; once the analysis focuses on within-country variation, the positive association does not persist as a robust result. Second, the temporal analyses do not provide consistent evidence that populism systematically precedes higher corruption, or that corruption systematically precedes more populist government. Third, decomposing the outcome by institutional sphere reveals that the contemporaneous negative association is concentrated in judicial corruption — the only dimension that survives FDR correction — while executive corruption shows no association; a possible positive delayed effect on legislative corruption emerges in some specifications but does not survive first-differencing or electoral-sample restrictions, and should be treated as suggestive rather than conclusive.

The main strength of the study is its sequential design. The report does not rely on a single correlation, but separates descriptive evidence, preprocessing choices, fixed-effects models, temporal diagnostics, robustness checks, and corruption-dimension comparisons. This makes the interpretation more cautious and better aligned with the research question. The use of V-Dem and V-Party also allows broad cross-national coverage and connects party-level populism to country-year corruption outcomes. At the same time, the study remains limited by the nature of the data and by the observational design. Populism and corruption are both expert-coded constructs, party information must be carried forward between elections, some institutional indicators have limited within-country variation, and the panel ends in 2019. These features make the evidence useful for comparative analysis, but they do not allow strong causal claims.

Future research could improve the design by exploiting more clearly exogenous changes in government, modelling election timing and duration in office more explicitly, distinguishing between different families of populist parties, and combining expert-coded indicators with administrative, judicial, or case-based corruption data. Overall, the results do not support a simple general claim that populist government systematically increases political corruption. Any relationship appears likely to be context-dependent and institutionally heterogeneous, although the available evidence remains fragile.

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A Appendix

Abbreviations, variable codes, and terms used across this project (code, notebooks, specification, and report). Items are grouped by kind. Variable definitions follow the V-Dem v16 and V-Party V2 codebooks.

A.1 Datasets and sources

Term	Meaning
V-Dem	Varieties of Democracy, here the **Country-Year: Full+Others (v16)** file (annual, country level). Outcomes and controls.
V-Party	V-Dem's **Country-Party-Date (V2)** file (party level, election-dated). Source of the populism predictor.
panel	The merged regression-ready country-year table, `data/processed/panel_populism_corruption.parquet`.

A.2 Outcome variables (V-Dem corruption family)

All variables run **higher = MORE corrupt** unless noted otherwise.

Code	Meaning
`v2x_corr`	**Political corruption index** (composite, 0–1). Equal-weighted average of the four spheres below. Primary outcome.
`v2x_execorr`	Executive corruption index (0–1). Bribery and embezzlement by the executive.
`v2x_pubcorr`	Public-sector corruption index (0–1). Bribery and theft by the bureaucracy.

Code	Meaning
`v2lgcrrpt`	Legislature corrupt activities. **Latent estimate runs higher = CLEANER** (reversed).
`v2jucorrdc`	Judicial corruption decision. **Latent estimate runs higher = CLEANER** (reversed).
`v2lgcrrpt_01`, `v2jucorrdc_01`	**Project-specific** orientation-harmonised 0–1 versions of the two above, built as `(4 - _ord) / 4` so higher = more corrupt. Coarse (5 levels); for plotting/descriptive use, not regression.
`v2ex...` (`v2exbribe`, `v2exembez`, `v2excrptps`, `v2exthftps`)	Component C-variables that build the executive and public indices.

A.3 Predictor and V-Party variables

Code	Meaning
`v2xpa_popul`	**Populism index** (party level, 0–1, higher = more populist). Built from anti-elitism and people-centrism.
`populism_governing`	**Primary predictor:** `v2xpa_popul` of the **senior governing party only** (`v2pagovsup==0`), forward-filled between elections. Vote-share weighting applies only when a country-year has more than one senior-governing-party record. Junior coalition partners and other parties enter only the *sensitivity* aggregations, not this measure.
`v2xpa_antiplural`	Anti-pluralism index (alternative predictor).
`v2pagovsup`	Government-support status of a party (`0` = senior governing party / head of government).
`v2pavote`	Party vote share (used as aggregation weight).
`v2paenname`	Party name (English).
`v2paanteli`, `v2papeople`	Anti-elitism, people-centrism, the C-variable components of `v2xpa_popul`.
`v2paculsup`	Cultural superiority / nativism (nationalism proxy).

A.4 Controls and keys

Code	Meaning
`v2x_polyarchy`	Electoral democracy index (0–1, higher = more democratic). Control / moderator.
`log_gdppc`	Natural log of `e_gdppc` (GDP per capita), the income control.
`country_id`, `year`	Join keys / panel index (numeric country id; calendar year).
`e_regiongeo`	UN-geoscheme region code (19 categories); used to stratify CV folds.
`e_pop`, `e_gdppc`	Population, GDP per capita (V-Dem external / Type-E variables).

A.5 Measurement-model suffixes and type prefixes (V-Dem)

Token	Meaning
`v2x_...`	**D-type** index, a composite (continuous, usually 0–1). The aggregation rule varies: some are built by Bayesian factor analysis, but many are formula-based, e.g. `v2x_corr` is a simple **average** of its four corruption spheres, and V-Party's `v2xpa_popul` is a **harmonic mean** of its components.
`v2...` (no `x`)	**C-type** variable, directly expert-coded (point estimate is a latent IRT score).
`e_...`	**E-type**, external / "Others" data (GDP, population, region).
`_ord`	Ordinal version (the original 0–4 category, e.g. for legislative/judicial corruption).
`_osp`	Ordinal-scale posterior prediction, the latent estimate mapped back onto the variable's **original ordinal scale** (commonly 0–4, *not* 0–1), for interpretation.
`_codelow` / `_codehigh`	Lower / upper bound of the measurement-uncertainty interval.
`_sd`	Posterior standard deviation (measurement uncertainty).
`_nr`	Number of coders / ratings for that cell; observations with `_nr` $\leq$ `3` are filtered.
`_mean`	Mean of the coder ratings (pre-IRT).
`_01`	**Project-specific:** orientation-harmonised 0–1 rescale (see corruption table above).
`_lag{1,2,3,5}` / `_lead{1,2,3}`	Within-country shift back / forward k years, used in the temporal direction analysis.

A.6 Statistical and methodological terms

Term	Meaning
FE	Fixed effects, absorb each entity's (or year's) own baseline as a free parameter.
RE	Random effects, treat baselines as draws uncorrelated with the regressors (a stronger assumption).
TWFE	Two-way fixed effects (country **and** year FE together).
OLS	Ordinary least squares. **PooledOLS / BetweenOLS** = pooled / between-country estimators.
within-R^2	Share of *within-country* variation explained (the relevant fit for FE); distinct from between/overall R^2.
within / between	The two parts of a variable's variance: movement of a country over time vs. differences across countries.
clustered SE	Standard errors that allow correlation within a cluster (here, a country) across years.
CI / SE	Confidence interval / standard error.
Hausman test	Diagnostic comparing FE vs. RE coefficients. Here the covariance difference is **not positive-definite**, so the statistic has no dependable chi-square interpretation and is **not** used as a formal rejection; FE is chosen from the research question, not this test.
Nickell bias	Small downward bias when a lagged outcome sits alongside entity FE; negligible at large T ($\sim$50 years here).

Term	Meaning
IRT	Item response theory, the latent-trait model V-Dem uses to turn coder ratings into scores.
BFA	Bayesian factor analysis, used to build *some* V-Dem `v2x_...` composites, but not all: several (including `v2x_corr`) are formula-based aggregates (averages), and V-Party's populism index uses a harmonic mean.
CV	Cross-validation. **GroupKFold / StratifiedGroupKFold** = CV that holds out whole groups (countries), optionally stratified (by region). **Nested CV** = an inner CV tunes hyperparameters, an outer CV scores, so tuning never sees the test data.
OOF	Out-of-fold, predictions/scores on held-out CV folds (honest out-of-sample R^2).
First-difference	Removes country effects by subtracting consecutive years ($\Delta y \sim \Delta x$) instead of demeaning, an alternative FE identification used as a robustness check.
Driscoll–Kraay	A standard-error estimator robust to cross-sectional dependence (e.g. regional/global corruption waves), used as an SE robustness variant.
Robustness ensemble	Re-running the same model under alternative estimators to show a conclusion is *stable*, as opposed to a leaderboard that picks the best-fitting model (which is avoided in this study).
SHAP	SHapley Additive exPlanations, per-feature contribution to a model's predictions (importance, **not** significance).
XGBoost	Gradient-boosted decision trees (`XGBRegressor`), used as the cross-national predictive check.

A.7 Project-specific terms

Term	Meaning
M1–M4	The nested two-way-FE specifications: M1 populism only; M2 +log GDP; M3 +polyarchy; M4 +lagged corruption. M1–M3 are the stability set; M4 is the separate short-run (dynamic) spec.
stability sample	The rows non-missing on the **M1–M3** variables (populism, log GDP, polyarchy); the three static models share it so coefficient movement reflects controls, not a shifting sample. M4 uses its **own** complete-case sample (it additionally needs the lag).
persistence baseline	Country-FE regression of corruption on last year's corruption, **documents how sticky corruption is**, not a performance bar the static models must beat (it is a different, dynamic specification/estimand).
between-vs-within divergence	Point-estimate pattern: **positive** across countries (pooled/between) but **negative** within countries (FE). With dependence-adjusted SEs the single-predictor estimates are imprecise; the within-negative is significant only once GDP/democracy are controlled (M2/M3).